IGCSE 0607 Practice Questions — E10.7 — Combined Questions & Markscheme

Scatter diagrams

Total Questions: 3 — **Total Marks:** 19 Marks

Table of Contents & Quick Navigation

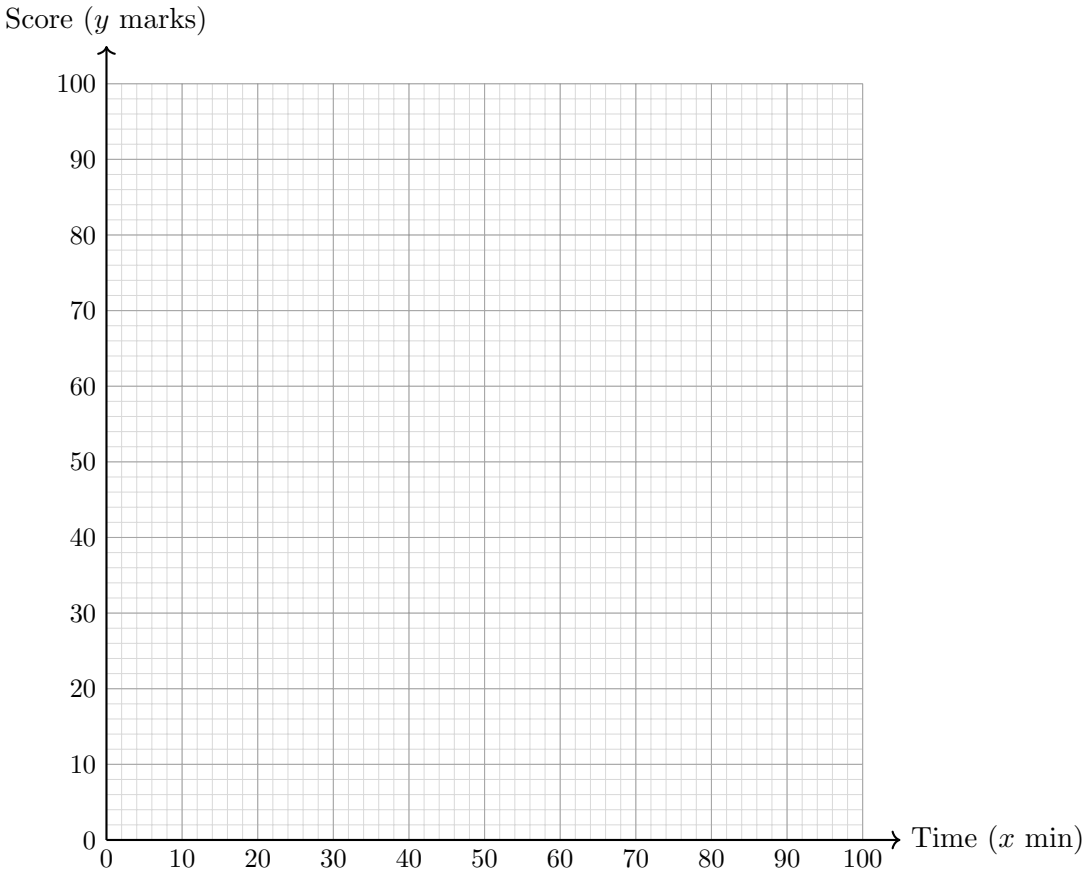
- **E10.7** Scatter diagrams
  → **Paper 4 (Calculator)** *(3 Qs)*

---

1 A teacher records the time, x minutes, spent studying and the score, y marks, for 8 students.
The table shows the results.

Time (x min)	10	20	30	40	50	60	70	80
Score (y marks)	15	25	20	40	50	70	60	80

(a) On the grid, complete the scatter diagram to show this information.



[2]

(b) What type of correlation is shown by the scatter diagram?

..... **[1]**

(c) On the scatter diagram, draw a line of best fit. **[1]**

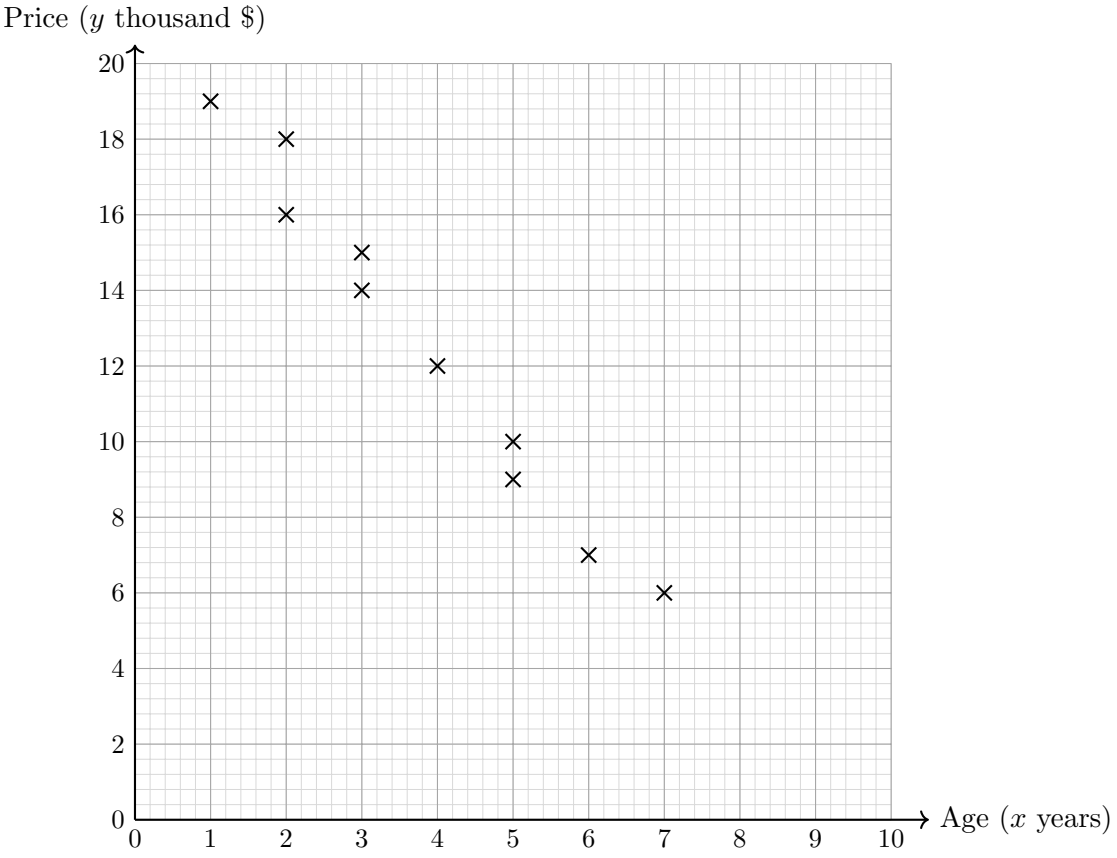
(d) Another student spent 45 minutes studying.
Use your line of best fit to estimate their score.

..... **[1]**

2 A car dealer records the age, x years, and the price, y thousand dollars, of 12 used cars of the same model.
The table shows the data.

Age (x years)	1	2	2	3	3	4	5	5	6	7	7	9
Price (y thousand \$)	19	18	16	15	14	12	10	9	7	6	5	1

(a) The scatter diagram shows the first 10 points from the table.
Plot the remaining 2 points on the scatter diagram.



[1]

(b) Calculate the mean age and the mean price.

Mean age = ... , Mean price = ... **[2]**

(c) On the scatter diagram, plot the mean point and draw a line of best fit. **[2]**

(d) State the type of correlation shown.

..... **[1]**

(e) Use your line of best fit to estimate the price of a car that is 8 years old.

..... thousand dollars **[1]**

3 The table shows the temperature, x°C, and the number of ice creams sold, y, at a shop over 8 days.

Temperature (x°C)	12	15	18	20	22	25	28	30
Ice creams sold (y)	40	52	68	75	84	98	110	115

(a) Calculate the mean temperature and the mean number of ice creams sold.

Mean temperature = ... , Mean ice creams sold = ... **[2]**

(b) Use a graphic display calculator to find the equation of the linear regression line for this data. Give your answer in the form $y = mx + c$, with values of m and c correct to 3 significant figures.

$y =$ ..... **[2]**

(c) The shop owner predicts the number of ice creams sold on a day when the temperature is 24°C. Use your equation to estimate this number.

..... **[2]**

(d) Comment on the reliability of using your equation to predict the number of ice creams sold on a day when the temperature is 42°C.

..... **[1]**

1 **(a)** All 8 points plotted correctly **[B2]**

(5, 6, or 7 points plotted correctly [B1])

(b) Positive **[B1]**

(c) Ruled line passing through the mean point $(45, 45)$ and fitting the trend of the data **[B1]**

(d) Value read correctly from their line of best fit (typically $42 - 48$) **[B1]**

[Back to Question] *[Total Marks: 5]*

2 **(a)** Points $(7, 5)$ and $(9, 1)$ correctly plotted **[B1]**

(b) Mean age $= 4.5$ **[B1]**

Mean price $= 11$ **[B1]**

(c) Mean point $(4.5, 11)$ plotted accurately **[B1]**

Ruled line of best fit passing through $(4.5, 11)$ **[B1]**

(d) Negative (or strong negative) **[B1]**

(e) Value read correctly from their line of best fit (typically $2 - 4$) **[B1]**

[Back to Question] *[Total Marks: 7]*

3 **(a)** Mean temperature $= 21.25$ **[B1]**

Mean ice creams sold $= 80.25$ **[B1]**

(b) $m = 4.28$ **[B1]**

$c = -10.6$ **[B1]**

(c) $y = 4.28 \times 24 - 10.6$ **[M1]**

92 (or 92.12) **[A1]**

(d) Unreliable because 42°C is outside the range of the given data (extrapolation) **[B1]**

[Back to Question] *[Total Marks: 7]*